設定新的Cisco Nexus 2300YC交換器

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設定新的Cisco Nexus 2300YC交換器

設定新的Cisco Nexus 2300YC交換器

您可以完成本章所述的步驟、設定新的Nexus 2300YC交換器。

在執行ONTAP __LW_9.6及更新版本的系統上安裝Nexus 2300YC交換器、首先設定IP位址和組態、讓交換器能夠透過管理介面進行通訊。然後您可以安裝NX-OS軟體和參考組態檔（RCF）。此程序旨在在準備Nexus 2300YC交換器、然後再新增控制器。

本程序中的範例使用下列交換器和節點命名法：

- Nexus 92300YC交換器名稱為「CS1」和「CS2」。
- 此程序中使用的範例會在第二個交換器上開始升級、即：* *CS2**。
- 叢集LIF名稱為節點1的「node1_clus1」和「node1_clus2」、節點2的「node2_clus1」和「node2_clus2」。
- IPspace名稱為「叢集」。
- 「cluster1：*：>」提示會指出叢集的名稱。
- 每個節點上的叢集連接埠名稱為「e0a」和「e0b」。

  請參閱 "*SUR1_* Hardware Universe" 以取得平台所支援的實際叢集連接埠。

- Nexus 2300YC交換器支援的交換器間連結（ISL）為連接埠1/65和1/66。
- Nexus 2300YC交換器支援的節點連線為連接埠1/1到1/66。
- 本程序中的範例使用兩個節點、但叢集中最多可有24個節點。

Cisco Nexus 92300YC交換器的初始安裝

您可以使用此程序來執行Cisco Nexus 2300YC交換器的初始安裝。

您可以從NetApp支援網站下載適用於交換器的NetApp Cisco NX-OS軟體、網址為： "mysupport.netapp.com"

NX-OS是Cisco Systems提供的Nexus系列乙太網路交換器和MDS系列光纖通道（FC）儲存區域網路交換器的網路作業系統。

本程序提供安裝交換器並讓其執行的程序摘要：

步驟

1. 將序列連接埠連接至您選擇的主機或序列連接埠。
2. 將管理連接埠（位於交換器的非連接埠端）連接至SFTP伺服器所在的相同網路。
3. 在主控台設定主機端序列設定：
   - 9600傳輸速率
   - 8個資料位元

- 1停止位元
- 同位元檢查：無
- 流程控制：無

4. 第一次開機或清除執行組態後重新開機、Nexus 92300YC交換器會在開機週期中循環運作。輸入* yes *中止開機自動資源配置、以中斷此週期。接著您會看到系統管理員帳戶設定：

```
$ VDC-1 %$ %POAP-2-POAP_INFO:   - Abort Power On Auto Provisioning [yes -
continue with normal setup, skip - bypass password and basic
configuration, no - continue with Power On Auto Provisioning]
(yes/skip/no)[no]: *y*
Disabling POAP.......Disabling POAP
2019 Apr 10 00:36:17 switch %$ VDC-1 %$ poap: Rolling back, please wait...
(This may take 5-15 minutes)

          ---- System Admin Account Setup ----

Do you want to enforce secure password standard (yes/no) [y]:
```

1. 輸入* y*以強制執行安全密碼標準：

```
Do you want to enforce secure password standard (yes/no) [y]: y
```

2. 輸入並確認使用者管理的密碼：

```
Enter the password for "admin":
Confirm the password for "admin":
```

3. 進入「基本系統組態」對話方塊：

```
This setup utility will guide you through the basic configuration of
the system. Setup configures only enough connectivity for management
of the system.

Please register Cisco Nexus9000 Family devices promptly with your
supplier. Failure to register may affect response times for initial
service calls. Nexus9000 devices must be registered to receive
entitled support services.

Press Enter at anytime to skip a dialog. Use ctrl-c at anytime
to skip the remaining dialogs.

Would you like to enter the basic configuration dialog (yes/no):
```

4. 建立另一個登入帳戶：

```
Create another login account (yes/no) [n]:
```

5. 設定唯讀和讀寫SNMP社群字串：

```
Configure read-only SNMP community string (yes/no) [n]:

Configure read-write SNMP community string (yes/no) [n]:
```

6. 設定叢集交換器名稱：

```
Enter the switch name : cs2
```

7. 設定頻外管理介面：

```
Continue with Out-of-band (mgmt0) management configuration? (yes/no)
[y]: y

Mgmt0 IPv4 address : 172.22.133.216

Mgmt0 IPv4 netmask : 255.255.224.0

Configure the default gateway? (yes/no) [y]: y

IPv4 address of the default gateway : 172.22.128.1
```

8. 設定進階IP選項：

```
Configure advanced IP options? (yes/no) [n]: n
```

9. 設定遠端登入服務：

```
Enable the telnet service? (yes/no) [n]: n
```

10. 設定SSH服務和SSH金鑰：

```
Enable the ssh service? (yes/no) [y]: y

       Type of ssh key you would like to generate (dsa/rsa) [rsa]: rsa

       Number of rsa key bits <1024-2048> [1024]: 2048
```

11. 設定其他設定：

```
Configure the ntp server? (yes/no) [n]: n

  Configure default interface layer (L3/L2) [L2]: L2

  Configure default switchport interface state (shut/noshut) [noshut]:
noshut

  Configure CoPP system profile (strict/moderate/lenient/dense)
[strict]: strict
```

12. 確認交換器資訊並儲存組態：

```
Would you like to edit the configuration? (yes/no) [n]: n

Use this configuration and save it? (yes/no) [y]: y

[] 100%
Copy complete, now saving to disk (please wait)...
Copy complete.
```

安裝NX-OS軟體

您可以使用此程序在Nexus 2300YC交換器上安裝NX-OS軟體。

步驟

1. 將叢集交換器連接至管理網路。
2. 使用「ping」命令來驗證與裝載NX-OS軟體和RCF之伺服器的連線。

   此範例可驗證交換器是否能以IP位址172.19.2.1連至伺服器：

   ```
   cs2# ping 172.19.2.1
   Pinging 172.19.2.1 with 0 bytes of data:

   Reply From 172.19.2.1: icmp_seq = 0. time= 5910 usec.
   ```

3. 將NX-OS軟體和EPLD映像複製到Nexus 2300YC交換器。

```
cs2# copy sftp: bootflash: vrf management
Enter source filename: /code/nxos.9.2.2.bin
Enter hostname for the sftp server: 172.19.2.1
Enter username: user1

Outbound-ReKey for 172.19.2.1:22
Inbound-ReKey for 172.19.2.1:22
user1@172.19.2.1's password:
sftp> progress
Progress meter enabled
sftp> get   /code/nxos.9.2.2.bin  /bootflash/nxos.9.2.2.bin
/code/nxos.9.2.2.bin  100% 1261MB   9.3MB/s   02:15
sftp> exit
Copy complete, now saving to disk (please wait)...
Copy complete.


cs2# copy sftp: bootflash: vrf management
Enter source filename: /code/n9000-epld.9.2.2.img
Enter hostname for the sftp server: 172.19.2.1
Enter username: user1

Outbound-ReKey for 172.19.2.1:22
Inbound-ReKey for 172.19.2.1:22
user1@172.19.2.1's password:
sftp> progress
Progress meter enabled
sftp> get   /code/n9000-epld.9.2.2.img  /bootflash/n9000-epld.9.2.2.img
/code/n9000-epld.9.2.2.img  100%  161MB   9.5MB/s   00:16
sftp> exit
Copy complete, now saving to disk (please wait)...
Copy complete.
```

4. 驗證NX-OS軟體的執行版本：

```
cs2# show version
Cisco Nexus Operating System (NX-OS) Software
TAC support: http://www.cisco.com/tac
Copyright (C) 2002-2018, Cisco and/or its affiliates.
All rights reserved.
The copyrights to certain works contained in this software are
owned by other third parties and used and distributed under their own
licenses, such as open source.  This software is provided "as is," and
unless
otherwise stated, there is no warranty, express or implied, including
```

```
but not
limited to warranties of merchantability and fitness for a particular
purpose.
Certain components of this software are licensed under
the GNU General Public License (GPL) version 2.0 or
GNU General Public License (GPL) version 3.0  or the GNU
Lesser General Public License (LGPL) Version 2.1 or
Lesser General Public License (LGPL) Version 2.0.
A copy of each such license is available at
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http://opensource.org/licenses/gpl-3.0.html and
http://www.opensource.org/licenses/lgpl-2.1.php and
http://www.gnu.org/licenses/old-licenses/library.txt.

Software
  BIOS: version 05.31
  NXOS: version 9.2(1)
  BIOS compile time:  05/17/2018
  NXOS image file is: bootflash:///nxos.9.2.1.bin
  NXOS compile time:  7/17/2018 16:00:00 [07/18/2018 00:21:19]


Hardware
  cisco Nexus9000 C92300YC Chassis
  Intel(R) Xeon(R) CPU D-1526 @ 1.80GHz with 16337884 kB of memory.
  Processor Board ID FDO220329V5

  Device name: cs2
  bootflash:  115805356 kB
Kernel uptime is 0 day(s), 4 hour(s), 23 minute(s), 11 second(s)

Last reset at 271444 usecs after Wed Apr 10 00:25:32 2019
  Reason: Reset Requested by CLI command reload
  System version: 9.2(1)
  Service:

plugin
  Core Plugin, Ethernet Plugin

Active Package(s):

cs2#
```

5. 安裝NX-OS映像。

   安裝映像檔會在每次重新開機時載入映像檔。

```
cs2# install all nxos bootflash:nxos.9.2.2.bin

Installer will perform compatibility check first. Please wait.
Installer is forced disruptive

Verifying image bootflash:/nxos.9.2.2.bin for boot variable "nxos".
[] 100% -- SUCCESS

Verifying image type.
[] 100% -- SUCCESS

Preparing "nxos" version info using image bootflash:/nxos.9.2.2.bin.
[] 100% -- SUCCESS

Preparing "bios" version info using image bootflash:/nxos.9.2.2.bin.
[] 100% -- SUCCESS

Performing module support checks.
[] 100% -- SUCCESS

Notifying services about system upgrade.
[] 100% -- SUCCESS



Compatibility check is done:
Module  bootable       Impact     Install-type  Reason
------  --------  --------------- ------------  ------
  1       yes      disruptive         reset     default upgrade is not
hitless



Images will be upgraded according to following table:

Module   Image         Running-Version(pri:alt            New-Version
Upg-Required
------  --------  --------------------------------------
------------------  ------------
  1       nxos                                  9.2(1)
9.2(2)         yes
  1       bios    v05.31(05/17/2018):v05.28(01/18/2018)
v05.33(09/08/2018)       yes


Switch will be reloaded for disruptive upgrade.
```

```
Do you want to continue with the installation (y/n)?  [n] y


Install is in progress, please wait.

Performing runtime checks.
[] 100% -- SUCCESS

Setting boot variables.
[] 100% -- SUCCESS

Performing configuration copy.
[] 100% -- SUCCESS

Module 1: Refreshing compact flash and upgrading bios/loader/bootrom.
Warning: please do not remove or power off the module at this time.
[] 100% -- SUCCESS
2019 Apr 10 04:59:35 cs2 %$ VDC-1 %$ %VMAN-2-ACTIVATION_STATE:
Successfully deactivated virtual service 'guestshell+'

Finishing the upgrade, switch will reboot in 10 seconds.
```

6. 在交換器重新開機後、驗證新版本的NX-OS軟體：

《如何版本》

```
cs2# show version

Cisco Nexus Operating System (NX-OS) Software
TAC support: http://www.cisco.com/tac
Copyright (C) 2002-2018, Cisco and/or its affiliates.
All rights reserved.
The copyrights to certain works contained in this software are
owned by other third parties and used and distributed under their own
licenses, such as open source.  This software is provided "as is," and
unless
otherwise stated, there is no warranty, express or implied, including
but not
limited to warranties of merchantability and fitness for a particular
purpose.
Certain components of this software are licensed under
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GNU General Public License (GPL) version 3.0  or the GNU
Lesser General Public License (LGPL) Version 2.1 or
Lesser General Public License (LGPL) Version 2.0.
A copy of each such license is available at
```

```
http://www.opensource.org/licenses/gpl-2.0.php and
http://opensource.org/licenses/gpl-3.0.html and
http://www.opensource.org/licenses/lgpl-2.1.php and
http://www.gnu.org/licenses/old-licenses/library.txt.

Software
  BIOS: version 05.33
  NXOS: version 9.2(2)
  BIOS compile time:  09/08/2018
  NXOS image file is: bootflash:///nxos.9.2.2.bin
  NXOS compile time:  11/4/2018 21:00:00 [11/05/2018 06:11:06]


Hardware
  cisco Nexus9000 C92300YC Chassis
  Intel(R) Xeon(R) CPU D-1526 @ 1.80GHz with 16337884 kB of memory.
  Processor Board ID FDO220329V5

  Device name: cs2
  bootflash:  115805356 kB
  Kernel uptime is 0 day(s), 0 hour(s), 3 minute(s), 52 second(s)

Last reset at 182004 usecs after Wed Apr 10 04:59:48 2019
  Reason: Reset due to upgrade
  System version: 9.2(1)
  Service:

plugin
  Core Plugin, Ethernet Plugin

Active Package(s):
```

7. 升級EPLD映像、然後重新啟動交換器。

```
cs2# show version module 1 epld

EPLD Device                      Version
---------------------------------------
MI FPGA                           0x7
IO FPGA                           0x17
MI FPGA2                          0x2
GEM FPGA                          0x2
GEM FPGA                          0x2
GEM FPGA                          0x2
GEM FPGA                          0x2

cs2# install epld bootflash:n9000-epld.9.2.2.img module 1
Compatibility check:
Module        Type         Upgradable        Impact   Reason
------  -----------------  ----------    ----------   ------
     1            SUP           Yes       disruptive  Module Upgradable

Retrieving EPLD versions.... Please wait.
Images will be upgraded according to following table:
Module  Type   EPLD              Running-Version   New-Version  Upg-
Required
------  ----  -------------      ---------------   -----------
------------
     1   SUP  MI FPGA                   0x07        0x07            No
     1   SUP  IO FPGA                   0x17        0x19           Yes
     1   SUP  MI FPGA2                  0x02        0x02            No
The above modules require upgrade.
The switch will be reloaded at the end of the upgrade
Do you want to continue (y/n) ?  [n] y

Proceeding to upgrade Modules.

Starting Module 1 EPLD Upgrade

Module 1 : IO FPGA [Programming] : 100.00% (     64 of      64 sectors)
Module 1 EPLD upgrade is successful.
Module        Type  Upgrade-Result
------  ------------------  --------------
     1         SUP         Success


EPLDs upgraded.

Module 1 EPLD upgrade is successful.
```

8. 交換器重新開機後、再次登入、並確認新版的EPLD已成功載入。

```
cs2# show version module 1 epld

EPLD Device                      Version
---- -------------------------- -------
MI   FPGA                        0x7
IO   FPGA                        0x19
MI   FPGA2                       0x2
GEM FPGA                         0x2
GEM FPGA                         0x2
GEM FPGA                         0x2
GEM FPGA                         0x2
```

安裝參考組態檔（RCF）

您可以在初次設定Nexus 2300YC交換器之後安裝RCF。

步驟

1. 將叢集交換器連接至管理網路。
2. 使用ping命令來驗證與裝載RCF的伺服器的連線能力。

   此範例可驗證交換器是否能以IP位址172.19.2.1連至伺服器：

```
cs2# ping 172.19.2.1
Pinging 172.19.2.1 with 0 bytes of data:

Reply From 172.19.2.1: icmp_seq = 0. time= 5910 usec.
```

3. 將RCF複製到Nexus 2300YC交換器：

```
cs2# copy sftp: bootflash: vrf management
Enter source filename: /code/Nexus_92300YC_RCF_v1.0.2.txt
Enter hostname for the sftp server: 172.19.2.1
Enter username: user1

Outbound-ReKey for 172.19.2.1:22
Inbound-ReKey for 172.19.2.1:22
user1@172.19.2.1's password:
sftp> progress
Progress meter enabled
sftp> get   /code/Nexus_92300YC_RCF_v1.0.2.txt
/bootflash/nxos.9.2.2.bin
/code/Nexus_92300YC_R  100% 9687   530.2KB/s   00:00
sftp> exit
Copy complete, now saving to disk (please wait)...
Copy complete.
```

4. 將RCF與交換器的執行組態合併：

```
cs2# copy bootflash:Nexus_92300YC_RCF_v1.0.2.txt running-config

Disabling ssh: as its enabled right now:
 generating ecdsa key(521 bits)......
generated ecdsa key

Enabling ssh: as it has been disabled
 this command enables edge port type (portfast) by default on all
interfaces. You
 should now disable edge port type (portfast) explicitly on switched
ports leading to hubs,
 switches and bridges as they may create temporary bridging loops.

Edge port type (portfast) should only be enabled on ports connected to a
single
 host. Connecting hubs, concentrators, switches, bridges, etc...  to
this
 interface when edge port type (portfast) is enabled, can cause
temporary bridging loops.
 Use with CAUTION

Edge Port Type (Portfast) has been configured on Ethernet1/1 but will
only
 have effect when the interface is in a non-trunking mode.

...

Copy complete, now saving to disk (please wait)...
Copy complete.
```

5. 在交換器上確認RCF已成功合併：

「如何執行設定」

```
cs2# show running-config
!Command: show running-config
!Running configuration last done at: Wed Apr 10 06:32:27 2019
!Time: Wed Apr 10 06:36:00 2019

version 9.2(2) Bios:version 05.33
switchname cs2
vdc cs2 id 1
  limit-resource vlan minimum 16 maximum 4094
  limit-resource vrf minimum 2 maximum 4096
  limit-resource port-channel minimum 0 maximum 511
  limit-resource u4route-mem minimum 248 maximum 248
  limit-resource u6route-mem minimum 96 maximum 96
  limit-resource m4route-mem minimum 58 maximum 58
  limit-resource m6route-mem minimum 8 maximum 8

feature lacp

no password strength-check
username admin password 5
$5$HY9Kk3F9$YdCZ8iQJ1RtoiEFa0sKP5IO/LNG1k9C4lSJfi5kesl
6  role network-admin
ssh key ecdsa 521

banner motd #

*
*
*  Nexus 92300YC Reference Configuration File (RCF) v1.0.2 (10-19-2018)
*
*
*
*  Ports 1/1  - 1/48: 10GbE Intra-Cluster Node Ports
*
*  Ports 1/49 - 1/64: 40/100GbE Intra-Cluster Node Ports
*
*  Ports 1/65 - 1/66: 40/100GbE Intra-Cluster ISL Ports
*
*
*
```

6. 儲存執行中的組態、以便在您重新啟動交換器時將其變成啟動組態：

```
cs2# *copy running-config startup-config*

[#########################################] 100%
Copy complete, now saving to disk (please wait)...
Copy complete.
```

7. 若為ONTAP Section9.6P8及更新版本、請啟用CSHM ASUP記錄收集功能以收集交換器相關的記錄檔：「系統叢集交換器記錄設定密碼」和「系統叢集交換器記錄啟用資料收集」

```
cs2# system cluster-switch log setup-password
Output example required here
cs2# system cluster-switch log enable-collection
Output example required here too
```

8. 重新啟動交換器、並驗證執行中的組態是否正確：

   "重裝"

```
cs2# reload

This command will reboot the system. (y/n)?  [n] y
```

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